Advanced Algebra

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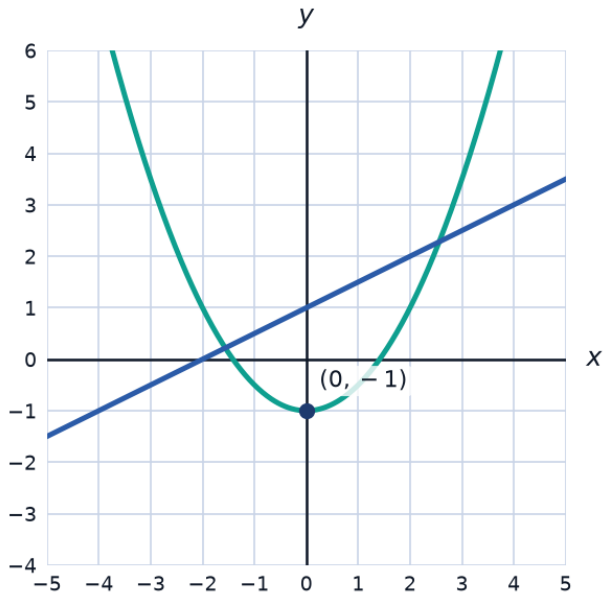
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Simplifying rational expressions

Simplify $\frac{x^2-4}{x^2+x-6}$.

2 [Foundational] • Target: Simplifying rational expressions

Simplify $\frac{x^2-5x+6}{x^2-9}$.



3 [Foundational] • Target: Simplifying rational expressions

Combine into a single fraction: $\frac{2}{x+1}+\frac{3}{x-1}$.

4 [Foundational] • Target: Simplifying rational expressions

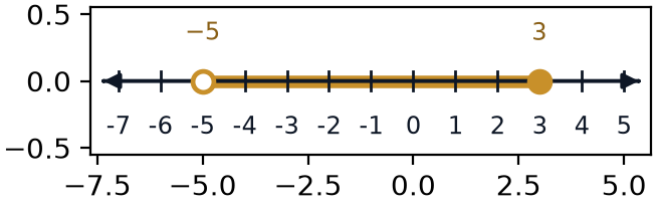
Simplify $\frac{3x^2-12}{x^2-4x+4}$.

5 [Foundational] • Target: Absolute value equations

Solve $|2x-5|=9$.

6 [Foundational] • Target: Absolute value equations

Solve $|3x+1|=|x-5|$.



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7 [Foundational] • Target: Absolute value equations

Solve $|x - 4| + 3 = 10$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Absolute value equations

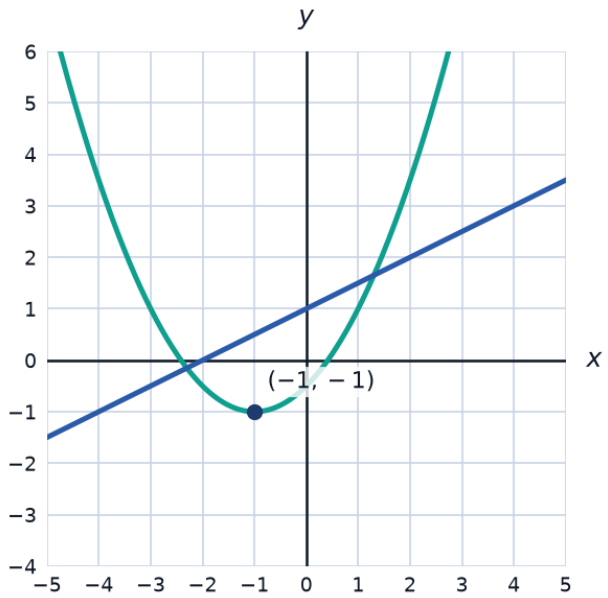
Solve $|4x + 3| = 11$.

9 [Intermediate] • Target: Polynomial division, remainder & factor theorems

Use the remainder theorem to find the remainder when $f(x) = x^3 + 2x^2 - 5x + 1$ is divided by $(x - 2)$.

10 [Intermediate] • Target: Polynomial division, remainder & factor theorems

Given $f(x) = x^3 + 3x^2 - 4x - 12$:



a Use the factor theorem to show that $(x + 3)$ is a factor of $f(x)$.

b Hence factor $f(x)$ completely.

11 [Intermediate] • Target: Polynomial division, remainder & factor theorems

Divide $x^3 - 2x^2 - 5x + 6$ by $(x - 3)$ using polynomial long division. State the quotient and remainder.

12 [Intermediate] • Target: Logarithmic equations

Solve $\log_3(x + 5) = 2$.

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13 [Intermediate] • Target: Logarithmic equations

Solve $\log_2(x+1) + \log_2(x-1) = 3$.

14 [Intermediate] • Target: Logarithmic equations

Solve $2\log_5 x = \log_5 16$.

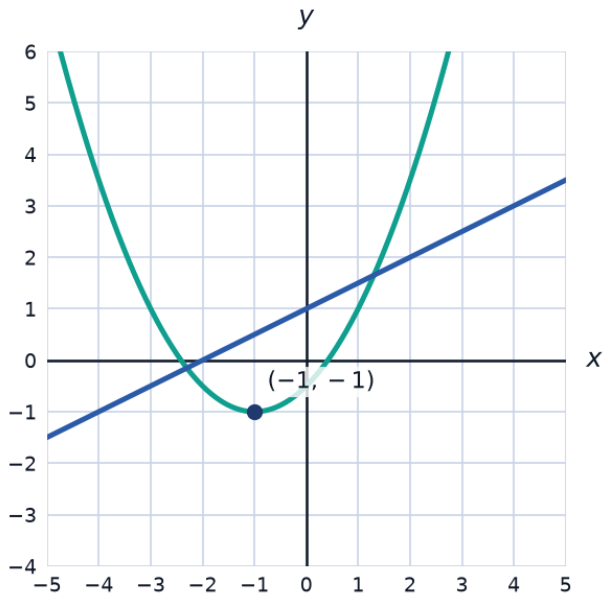
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Logarithmic equations • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Solve $\log_3(x-2) = 4$.

16 [Challenging] • Target: Exponential equations • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Solve $3^{2x-1} = 27$.



17 [Challenging] • Target: Exponential equations • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Solve $5^{x+2} = 5^{3x-4}$.

18 [Challenging] • Target: Exponential equations • Cross-Lesson: 5.01 Trigonometry

Solve $2^x = 15$ for x, giving your answer to three decimal places.

19 [Challenging] • Target: Exponential equations • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Solve $4^x = 2^{x+3}$.

20 [Challenging] • Target: Nonlinear systems of equations • Cross-Lesson: 7.01 Solid Geometry & Similarity

Solve the system $y = x^2 - 1$ and $y = 2x + 2$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Simplifying rational expressions

Solve $2^{2x-1} = 128$.

a. 5

b. 7

c. 3

d. 4

22 [Foundational] • Target: Absolute value equations

Solve $2^{2x-1} = 2$.

a. 0

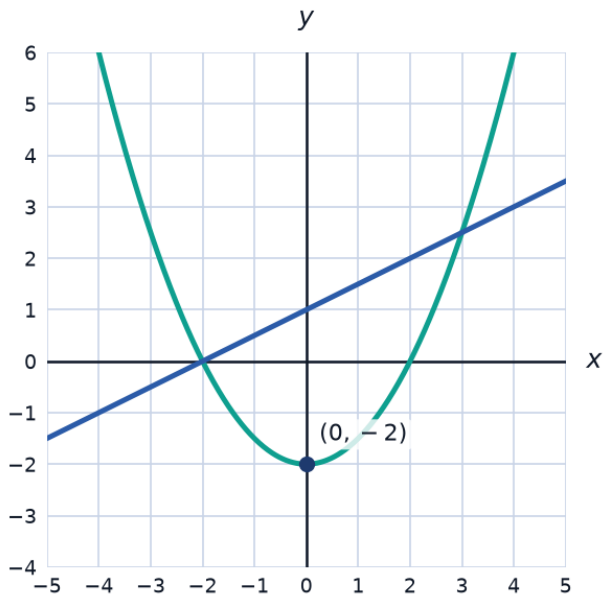
b. 1

c. 3

d. 2

23 [Foundational] • Target: Polynomial division, remainder & factor theorems

Solve $2^{2x-1} = 32$.



a. 5

b. 3

c. 2

d. 4

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24 [Foundational] • Target: Logarithmic equations

Solve $3^{2x-1} = 2187$.

a. 3

b. 4

c. 7

d. 5

25 [Foundational] • Target: Exponential equations

Solve $7^{2x-1} = 16807$.

a. 4

b. 2

c. 5

d. 3

26 [Foundational] • Target: Nonlinear systems of equations

Solve $5^{2x-1} = 48828125$.

a. 11

b. 5

c. 6

d. 7

27 [Foundational] • Target: Rational equations and extraneous solutions

Solve $5^{2x-1} = 125$.

a. 1

b. 3

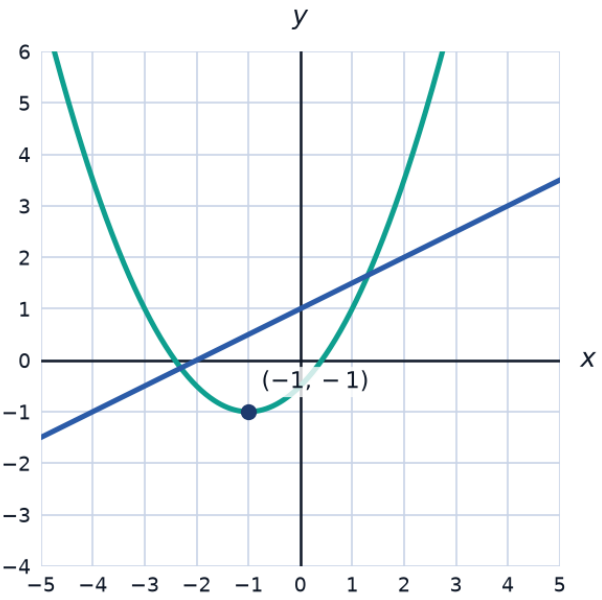
c. 4

d. 2

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28 [Foundational] • Target: Systems of inequalities and linear programming

Use the remainder theorem to find the remainder when $f(x) = x^3 + 5x^2 - x - 5$ is divided by $(x - 1)$.



a. 0

b. -5

c. -2

d. 2

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: production planning at a furniture workshop

Use the remainder theorem to find the remainder when $f(x) = x^3 - x^2 - 2x + 2$ is divided by $(x - (-4))$.

a. -70

b. -54

c. -72

d. -68

30 [Intermediate] • Target: Polynomial Theorems & Synthetic Division

Use the remainder theorem to find the remainder when $f(x) = x^3 + x^2 - 7x + 2$ is divided by $(x - (-4))$.

a. -16

b. -18

c. -34

d. -20

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31 [Intermediate] • Target: Rational Equations & Partial Fractions

Use the remainder theorem to find the remainder when $f(x) = x^3 - x^2 + 7x - 9$ is divided by $(x - 5)$.

a. 151

b. 128

c. 126

d. 124

32 [Intermediate] • Target: Radical Equations & Fractional Exponents

Use the remainder theorem to find the remainder when $f(x) = x^3 - 4x - 4$ is divided by $(x - (-1))$.

a. 1

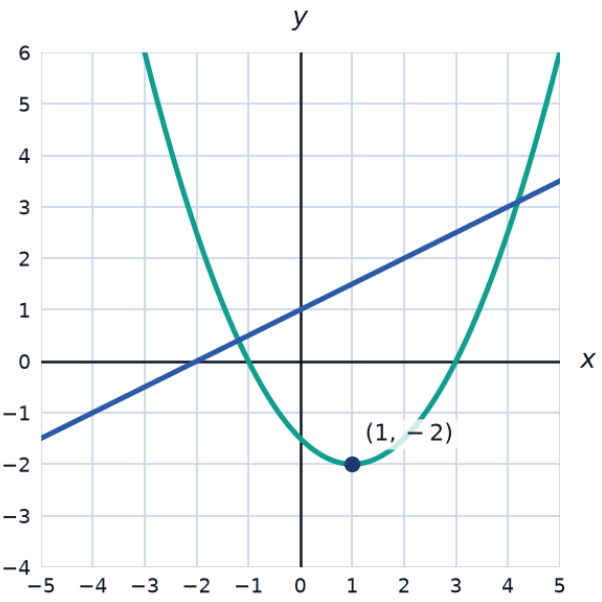
b. 2

c. -3

d. -1

33 [Intermediate] • Target: Non-Linear Systems & Inequalities

Use the remainder theorem to find the remainder when $f(x) = x^3 - 3x^2 + 6x + 1$ is divided by $(x - (-1))$.



a. -7

b. -9

c. -11

d. -6

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34 [Intermediate] • Target: Simplifying rational expressions

Solve $\log_2(x+3) + \log_2(x-3) = 4$.

a. 7

b. 3

c. -5

d. 5

35 [Intermediate] • Target: Absolute value equations

Solve $\log_2(x+1) + \log_2(x-1) = 3$.

a. -3

b. 5

c. 3

d. 1

36 [Intermediate] • Target: Polynomial division, remainder & factor theorems

Solve $\log_3(x+4) + \log_3(x-4) = 2$.

a. 7

b. -5

c. 3

d. 5

37 [Intermediate] • Target: Logarithmic equations

Solve $5^x = 141$ for x, giving your answer to three decimal places.

a. 2.575

b. 4.075

c. 3.075

d. 3.575

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Exponential equations • Cross-Lesson: 5.01 Trigonometry

Solve $5^x = 410$ for x, giving your answer to three decimal places.

a. 3.738

b. 4.238

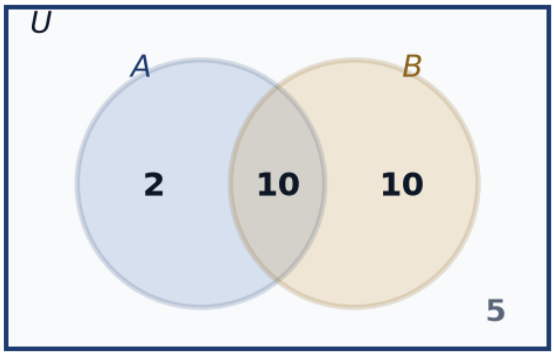
c. 4.738

d. 3.238

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39 [Challenging] • Target: Nonlinear systems of equations • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Solve the system $y = x^2 + x + 10$ and $y = -5x + 10$. Find the x-coordinates of the intersection points.



a. $-4, 0$

b. $6, 0$

c. $-6, 0$

d. $-6, -2$

40 [Challenging] • Target: Rational equations and extraneous solutions • Cross-Lesson: 7.01 Solid Geometry & Similarity

Solve the system $y = x^2 + 13x + 27$ and $y = 4x + 7$. Find the x-coordinates of the intersection points.

a. $-5, -4$

b. $5, 4$

c. $-3, -4$

d. $-5, -6$

41 [Challenging] • Target: Polynomial Theorems & Synthetic Division • Cross-Lesson: 8.01 Statistics & Probability

If $(x - 2)$ is a factor of $P(x) = 2x^3 - 3x^2 + kx - 10$, what is the value of k?

a. 1

b. 2

c. 3

d. 4

42 [Challenging] • Target: Rational Equations & Partial Fractions • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Solve the rational equation: $\frac{x}{x-2} + \frac{2}{x+2} = \frac{8}{x^2-4}$.

a. $x = 2$

b. $x = -2$

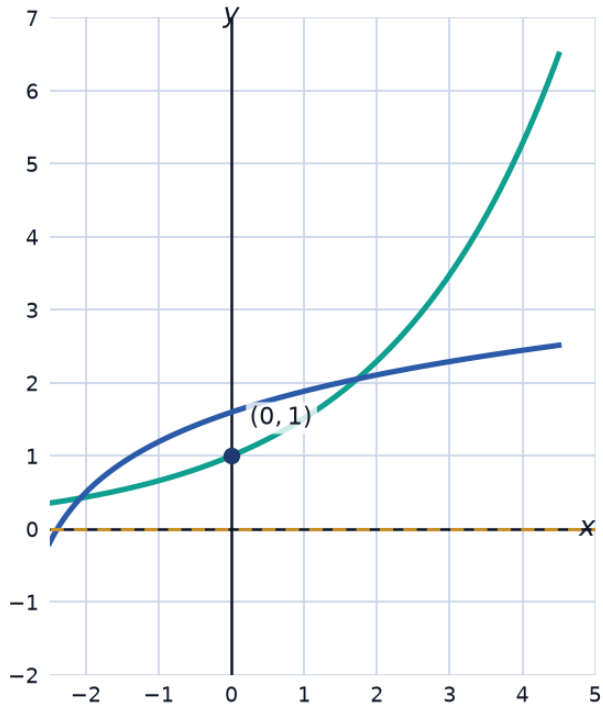
c. $x = 2$ and $x = -2$

d. No real solution

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43 [Challenging] • Target: Rational Equations & Partial Fractions • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Simplify the compound fraction: $\frac{\frac{1}{x+h}-\frac{1}{x}}{h}$ for $h \neq 0$.



a. $-\frac{1}{x(x+h)}$

b. $\frac{1}{x(x+h)}$

c. $-\frac{1}{x^2}$

d. $\frac{h}{x(x+h)}$

44 [Challenging] • Target: Radical Equations & Fractional Exponents • Cross-Lesson: 4.01 Lines, Triangles & Polygons

If $x^{2/3} - 5x^{1/3} + 6 = 0$, what are all real values of x?

a. 8 and 27

b. 2 and 3

c. ± 8

d. ± 27

45 [Challenging] • Target: Non-Linear Systems & Inequalities • Cross-Lesson: 5.01 Trigonometry

Solve the non-linear system: $y = x^2 - 2x - 3$ and $y = 2x + 2$. What is the sum of the x-coordinates of the intersection points?

a. 2

b. 4

c. -4

d. 6

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